

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. BPE-602: Elementary Nanoscience and Exotic Materials

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer in brief any **four** of the following: [4 × 3.5 = 14]

- (a) Explain how the band gap of a material changes with a change in its dimensions.
- (b) How is the crystallite size determined using X-ray diffraction (XRD) patterns?
- (c) Explain the Fibonacci sequence.
- (d) What is meant by surface plasmon resonance? How is it related to nanomaterials?
- (e) What are Type-I and Type-II superconductors?

Question 2.

- (a) What are the properties of high- T_c superconductors? Explain how they are different from conventional superconductors. [8]
- (b) What is the Meissner effect? A superconducting lead has a critical temperature of 7.26 K at zero magnetic field and a critical field of 8×10^5 A/m at 0 K. Find the critical field at 5 K. [6]

Question 3.

- (a) What are colossal magnetoresistance (CMR) materials? Discuss the double exchange mechanism of colossal magnetoresistance. [8]
- (b) How is CMR different from ordinary resistance and Giant Magnetoresistance (GMR)? [6]

OR

- (a) What are carbon nanotubes (CNTs)? Describe the armchair, zigzag, and chiral types of CNTs using suitable diagrams. [8]
- (b) Write in brief the working of a transmission electron microscope (TEM). [6]

Question 4.

- (a) Explain the particle size effect on the optical spectra of nanomaterials. Explain how the optical spectra can be used to determine the size of nanoparticles. [9]
 - (b) Explain how the band gap of nanoparticles is evaluated using their UV-Vis absorption spectra. [5]
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Question 5.

- (a) What are quasicrystals? Give one example. Provide the construction of a 1D quasicrystal by projecting a 2D periodic lattice onto a line with an irrational slope. [8]
- (b) What is a Quantum Dot (QD)? If the size of a QD is halved, what will happen to the frequency of the fluorescent light emitted by it? [6]

OR

- (a) Describe the differences between the physical and chemical synthesis routes of nanomaterials. Explain the working principle and limitations of high-energy ball milling. [9]
- (b) Describe the bio-route of synthesis of nanoparticles. [5]